

ANATOMY - MOCK MIDTERM 2

Head and neck: the face, the mouth and nose, and the eye

Instructions. 24 questions. Only **one answer** is correct for each question. Suggested time: 40 minutes. Answer on your own first, then check the answer key at the end, which explains each answer and names the trap the question uses. This paper was written for practice by Sina and is not an official exam.

Question 1/24 Which anatomical parts form the roof of the orbit?

- A. The greater wing of the sphenoid bone and the zygomatic bone
- B. The orbital plate of the frontal bone and the lesser wing of the sphenoid bone
- C. The orbital surface of the maxillary bone
- D. The cribriform plate of the ethmoid bone

Question 2/24 Regarding the ethmoid bone:

- A. The inferior nasal concha is part of it
- B. The cribriform plate is pierced by the optic nerve
- C. It is exclusively a facial bone
- D. The crista galli gives attachment to the falx cerebri

Question 3/24 The temporal bone:

- A. Its mastoid process lies in front of the external acoustic meatus
- B. Its styloid process gives attachment to the masseter muscle
- C. The stylomastoid foramen transmits the facial nerve
- D. Its petrous part separates the anterior and middle cranial fossae

Question 4/24 Which opening belongs to the posterior cranial fossa?

- A. The optic canal
- B. The foramen ovale
- C. The foramen rotundum
- D. The jugular foramen

Question 5/24 In the Mandible:

- A. The coronoid process gives insertion to the temporalis muscle
- B. The submandibular fossa lies above the mylohyoid line
- C. The mental foramen is located on the medial surface of the ramus
- D. The condylar process lies in front of the mandibular notch

Question 6/24 The zygomatic arch is formed by:

- A. The zygomatic process of the temporal bone and the temporal process of the zygomatic bone
- B. Mainly the temporal bone with smaller contribution from the frontal bone
- C. Only the zygomatic bone
- D. The zygomatic process of the maxillary bone and the temporal process of the zygomatic bone

Question 7/24 Regarding the nasal turbinates (or conchas) and meatus:

- A. The inferior turbinate belongs to the ethmoid bone
- B. The nasolacrimal duct opens at the level of the inferior meatus
- C. The superior meatus lies below the middle turbinate
- D. The sphenoid sinus opens in the middle meatus

Question 8/24 Regarding the sphenoid sinus:

- A. It is related above to the pituitary gland
- B. It is located in the greater wing of the sphenoid
- C. The septum dividing it is always midline
- D. It drains in the middle meatus

Question 9/24 The oral cavity proper:

- A. Ends posteriorly at the laryngeal inlet
- B. Has the buccinator muscle as its floor
- C. Communicates with the vestibule behind the third molars when the jaws are closed
- D. Receives the opening of the parotid duct

Question 10/24 Taste from the anterior 2/3 of the tongue is carried by:

- A. The lingual nerve fibers of the mandibular nerve
- B. The chorda tympani of the facial nerve
- C. The hypoglossal nerve
- D. The glossopharyngeal nerve

Question 11/24 Regarding the palate:

- A. The hard palate is formed mainly by the vomer
- B. The palatine aponeurosis is the expanded tendon of the tensor veli palatini
- C. The tensor veli palatini is supplied by the pharyngeal plexus
- D. The levator veli palatini is supplied by the mandibular nerve

Question 12/24 The parotid gland:

- A. Has a duct that pierces the masseter muscle
- B. Is a mixed gland with mucous predominance
- C. Is the smallest salivary gland
- D. Is crossed by the facial nerve within its substance

Question 13/24 Orbicularis oculi:

- A. Is supplied by the oculomotor nerve
- B. Raises the upper eyelid
- C. Its palpebral part closes the eyelids forcefully
- D. Its palpebral part closes the eyelids gently, as in blinking

Question 14/24 The masseter muscle:

- A. Is supplied by the facial nerve
- B. Inserts on the coronoid process only
- C. Arises from the zygomatic arch
- D. Depresses the mandible

Question 15/24 Which muscle of mastication helps to open the mouth?

- A. The masseter muscle
- B. The lateral pterygoid muscle
- C. The temporalis muscle
- D. The medial pterygoid muscle

Question 16/24 The external carotid artery:

- A. Gives off the ophthalmic artery
- B. Lies posterolateral to the internal carotid artery at its origin
- C. Arises at the level of the cricoid cartilage
- D. Terminates in the parotid gland behind the neck of the mandible

Question 17/24 The middle meningeal artery:

- A. Supplies the parotid gland
- B. Is a branch of the superficial temporal artery
- C. Has an anterior branch that lies under the pterion
- D. Enters the skull through the foramen ovale

Question 18/24 A patient squeezes a boil beside his nose. A few days later he develops fever, a protruding eye and paralysis of the eye movements. The infection has most likely spread through:

- A. The submental lymph nodes
- B. The external jugular vein
- C. The ophthalmic veins to the cavernous sinus
- D. The facial artery

Question 19/24 The superior orbital fissure transmits all of the following EXCEPT:

- A. The oculomotor nerve
- B. The optic nerve
- C. The trochlear nerve
- D. The abducens nerve

Question 20/24 The tear film:

- A. Has an oily layer produced by the goblet cells
- B. Is made of only two layers
- C. Has a mucin layer that attaches it to the eye
- D. Has a watery layer produced by the meibomian glands

Question 21/24 Regarding the cornea:

- A. It gives insertion to the extraocular muscles
- B. It is highly sensitive and supplied by the ophthalmic nerve
- C. It forms five sixths of the outer coat
- D. It is richly vascularized

Question 22/24 Accommodation for near vision:

- A. The lens becomes thicker and more convex
- B. The zonular fibers become taut
- C. The ciliary muscle relaxes
- D. Is mediated by sympathetic fibers

Question 23/24 A patient complains of vertical double vision, worse when reading or going down the stairs. The most likely nerve involved is:

- A. The trochlear nerve
- B. The abducens nerve
- C. The optic nerve
- D. The facial nerve

Question 24/24 An expanding mass of pituitary tumor directly compresses the optic chiasma from below. This will lead to which visual defect?

- A. Right homonymous hemianopia
- B. Loss of the nasal visual fields
- C. Blindness of one eye
- D. Bitemporal hemianopia

Answer key, mock midterm 2

Q	Ans	Why	Trap used
1	B	Roof = frontal + lesser wing; cribriform plate is the roof of the nose.	Cribriform plate distractor (tested in a past exam).
2	D	The inferior concha is an independent bone; the cribriform plate carries CN I; the ethmoid is counted as a cranial bone.	'Exclusively'; nerve swap.
3	C	Petrous part separates middle and posterior fossae; mastoid is behind the meatus; styloid = styloglossus, stylohyoid, stylopharyngeus.	Anterior/posterior swaps.
4	D	Posterior fossa: internal acoustic meatus, jugular foramen, hypoglossal canal, foramen magnum.	Three middle-fossa openings as distractors.
5	A	Mental foramen: outer body below the premolars; condyle is behind the notch; submandibular fossa is below the line.	Real-paper stem; front/behind and above/below swaps.
6	A	The two processes meet at the zygomaticotemporal suture.	Exact template from the real mock.
7	B	The inferior concha is independent; the superior meatus is under the superior concha; the sphenoid opens in the spheno-ethmoidal recess.	Real-paper stem.
8	A	It lies in the body; above it: hypophyseal fossa and optic chiasma; laterally: cavernous sinus.	'Always midline' appears on two real papers.
9	C	Floor = mylohyoid; ends at the oropharyngeal isthmus; parotid duct opens into the vestibule.	Part swap (vestibule vs cavity proper).
10	B	The lingual nerve carries general sensation; taste travels in the chorda tympani that runs with it.	General sensation vs taste.
11	B	Tensor = V3; all other palatine muscles = pharyngeal plexus; hard palate = maxilla + palatine.	Nerve swapped between the two veli palatini muscles.
12	D	Purely serous, the largest; its duct crosses the masseter and pierces the buccinator.	Crosses vs pierces.
13	D	Palpebral = gentle, involuntary; orbital = forceful, voluntary; VII. Levator raises the lid.	Two near-identical options; read every word.
14	C	Inserts on the angle and lateral ramus; elevates; V3.	'Only'.
15	B	The lateral pterygoid pulls the condyle and disc forward; the other three elevate.	-
16	D	Bifurcation at the upper border of the thyroid cartilage (C3-C4); anteromedial at origin; ophthalmic = ICA.	Level and position swaps.
17	C	Branch of the maxillary; enters by the foramen spinosum.	Ovale/spinosum swap.
18	C	Danger area: facial/angular vein → ophthalmic veins → cavernous sinus (III, IV, VI there).	Clinical vignette.
19	B	The optic nerve uses the optic canal.	EXCEPT format, as in the orbital quiz.
20	C	Oil = meibomian; water = lacrimal gland; mucin = goblet cells; three layers.	Sources rotated between layers.
21	B	1/6, avascular, V1. The sclera (5/6) receives the muscles.	Cornea/sclera swap.

Q	Ans	Why	Trap used
22	A	CN III parasympathetic: ciliary muscle contracts, zonules relax, lens rounds up.	Direction trap (contract → zonules relax).
23	A	Superior oblique (IV) depresses the adducted eye.	Clinical vignette.
24	D	Crossing nasal fibres carry the temporal fields of both eyes.	Same stem as the real mock (minus the figure).